THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28	Perimeter $= 4\times$ side.
2	**(a)** $5n + 2$	Write the rule using a letter-number.
3	**(d)** 16	Combine the like terms first: $m + am = (1 + a)m$.
4	**(b)** $5x$	Subtract the coefficients of the like terms.
5	**(b)** 14	Add inside the bracket first.
6	**(c)** $4(x + 3)$	Use brackets to keep the order of operations.
7	**(b)** 110	Fare $=$ fixed charge $+$ rate $\times$ distance.
8	**(d)** 19	Substitute the value of n.
9	**(c)** 48	Perimeter $= 4\times$ side.
10	**(a)** $5n + 2$	Write the rule using a letter-number.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute n in $3n + 1$ (1); $3(5) + 1 = 16$ matchsticks (1).	1+1
12	Collect like terms: $(7 - 3)x$ and $(4 + 6)$ (1); Answer: $4x + 10$ (1).	1+1
13	Substitute: $5(6) - 3(4)$ (1); $= 18$ (1).	1+1
14	Substitute n in $3n + 1$ (1); $3(8) + 1 = 25$ matchsticks (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Figure 4: $3(4) + 2 = 14$ (1); Figure 5: $3(5) + 2 = 17$ (1); Difference $= 17 - 14 = 3$ (1).	1+1+1
16	Group the x-terms and the y-terms (1); x-terms: $5x + 3x = 8x$ (1); Sum $= 8x + 5y$ (1).	1+1+1
17	Father's age $= 3s + 2$ (1); At $s = 10$: $3(10) + 2$ (1); Father's age $= 32$ years (1).	1+1+1
	OR: Figure 6: $3(6) + 2 = 20$ (1); Figure 7: $3(7) + 2 = 23$ (1); Difference $= 23 - 20 = 3$ (1).	1+1+1
18	Group the x-terms and the y-terms (1); x-terms: $7x + 2x = 9x$ (1); Sum $= 9x + 5y$ (1).	1+1+1
	OR: Father's age $= 2s + 8$ (1); At $s = 14$: $2(14) + 8$ (1); Father's age $= 36$ years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Notebooks cost $12n$ and pens cost 20 (1); Total cost $= 12n + 20$ (1); At $n = 10$: 140 (1); Change $= 200 - 140 = 60$ (2).	1+1+1+2
	OR: Total $= 8r + 6$ (1); For $r = 12$: 102 (2); For $r = 14$: 118 (2).	1+2+2
20	Notebooks cost $15n$ and pens cost 24 (1); Total cost $= 15n + 24$ (1); At $n = 8$: 144 (1); Change $= 200 - 144 = 56$ (2).	1+1+1+2
	OR: Total $= 10r + 4$ (1); For $r = 9$: 94 (2); For $r = 11$: 114 (2).	1+2+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 18	Combine the like terms first: $m + am = (1 + a)m$.
2	**(d)** $7x$	Subtract the coefficients of the like terms.
3	**(a)** 21	Add inside the bracket first.
4	**(c)** $y + 2z$	Use brackets to keep the order of operations.
5	**(a)** 100	Fare = fixed charge + rate × distance.
6	**(b)** 28	Substitute the value of n.
7	**(c)** 36	Perimeter = $4\times$ side.
8	**(c)** $2n$	Write the rule using a letter-number.
9	**(a)** 25	Combine the like terms first: $m + am = (1 + a)m$.
10	**(c)** $7x$	Subtract the coefficients of the like terms.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Collect like terms: $(9 - 5)x$ and $(2 + 8)$ (1); Answer: $4x + 10$ (1).	1+1
12	Substitute: $4(9) - 7(2)$ (1); $= 22$ (1).	1+1
13	Substitute n in $3n + 1$ (1); $3(10) + 1 = 31$ matchsticks (1).	1+1
14	Collect like terms: $(12 - 7)x$ and $(5 + 3)$ (1); Answer: $5x + 8$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Figure 10: $3(10) + 2 = 32$ (1); Figure 11: $3(11) + 2 = 35$ (1); Difference $= 35 - 32 = 3$ (1).	1+1+1
16	Group the x-terms and the y-terms (1); x-terms: $4x + 6x = 10x$ (1); Sum $= 10x + 5y$ (1).	1+1+1
17	Father's age $= 3s + 5$ (1); At $s = 12$: $3(12) + 5$ (1); Father's age $= 41$ years (1).	1+1+1
	OR: Figure 8: $3(8) + 2 = 26$ (1); Figure 9: $3(9) + 2 = 29$ (1); Difference $= 29 - 26 = 3$ (1).	1+1+1
18	Group the x-terms and the y-terms (1); x-terms: $9x + x = 10x$ (1); Sum $= 10x + 3y$ (1).	1+1+1
	OR: Father's age $= 4s + 2$ (1); At $s = 9$: $4(9) + 2$ (1); Father's age $= 38$ years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Notebooks cost $9n$ and pens cost 30 (1); Total cost $= 9n + 30$ (1); At $n = 12$: 138 (1); Change $= 200 - 138 = 62$ (2).	1+1+1+2
	OR: Total $= 6r + 8$ (1); For $r = 15$: 98 (2); For $r = 17$: 110 (2).	1+2+2
20	Notebooks cost $20n$ and pens cost 24 (1); Total cost $= 20n + 24$ (1); At $n = 7$: 164 (1); Change $= 200 - 164 = 36$ (2).	1+1+1+2
	OR: Total $= 12r + 5$ (1); For $r = 10$: 125 (2); For $r = 12$: 149 (2).	1+2+2

Mistakes to avoid:

- Adding unlike terms: $4x + 3y = 7xy$ is wrong.
- Writing $3a + 2a = 5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x + 3) - (2x - 4) \neq 3x - 1$.
- Confusing 'n less than a' $(a - n)$ with $n - a$.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 28	Add inside the bracket first.
2	**(a)** $2(a-b)$	Use brackets to keep the order of operations.
3	**(c)** 140	Fare = fixed charge + rate × distance.
4	**(d)** 34	Substitute the value of n.
5	**(d)** 60	Perimeter $= 4\times$ side.
6	**(b)** $2n-1$	Write the rule using a letter-number.
7	**(a)** 24	Combine the like terms first: $m+am=(1+a)m$.
8	**(a)** $9x$	Subtract the coefficients of the like terms.
9	**(d)** 25	Add inside the bracket first.
10	**(c)** $pq+5$	Use brackets to keep the order of operations.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $8(5)-2(3)$ (1); $=34$ (1).	1+1
12	Substitute n in $3n+1$ (1); $3(12)+1=37$ matchsticks (1).	1+1
13	Collect like terms: $(10-4)x$ and $(9+1)$ (1); Answer: $6x+10$ (1).	1+1
14	Substitute: $6(7)-5(6)$ (1); $=12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Figure 5: $3(5)+2=17$ (1); Figure 6: $3(6)+2=20$ (1); Difference $=20-17=3$ (1).	1+1+1
16	Group the x-terms and the y-terms (1); x-terms: $2x+8x=10x$ (1); Sum $=10x+3y$ (1).	1+1+1
17	Father's age $=2s+6$ (1); At $s=16$: $2(16)+6$ (1); Father's age $=38$ years (1).	1+1+1
	OR: Figure 12: $3(12)+2=38$ (1); Figure 13: $3(13)+2=41$ (1); Difference $=41-38=3$ (1).	1+1+1
18	Group the x-terms and the y-terms (1); x-terms: $6x+5x=11x$ (1); Sum $=11x+y$ (1).	1+1+1
	OR: Father's age $=3s+4$ (1); At $s=11$: $3(11)+4$ (1); Father's age $=37$ years (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Notebooks cost $18n$ and pens cost 20 (1); Total cost $=18n+20$ (1); At $n=9$: 182 (1); Change $=200-182=18$ (2).	1+1+1+2
	OR: Total $=9r+3$ (1); For $r=11$: 102 (2); For $r=13$: 120 (2).	1+2+2
20	Notebooks cost $14n$ and pens cost 21 (1); Total cost $=14n+21$ (1); At $n=15$: 231 (1); Change $=300-231=69$ (2).	1+1+1+2
	OR: Total $=7r+9$ (1); For $r=14$: 107 (2); For $r=16$: 121 (2).	1+2+2

Mistakes to avoid:

- Adding unlike terms: $4x+3y=7xy$ is wrong.
- Writing $3a+2a=5a^2$; the letter part does not change when like terms are added.
- Writing $3n$ as '3 plus n'; it means 3 times n.
- Not changing all signs when subtracting an expression in brackets: $(5x+3)-(2x-4)\neq 3x-1$.
- Confusing 'n less than a' $(a-n)$ with $n-a$.

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